H4 H4_phaseavg environmental-variable selection

STEP 1
Inventory (229)

UCM

- AIR RH
- AIR T mid
- HUMIDEX
- MRT
- PET
- SND dBA
- SUN Gh
- UTCI
- WIND AWS
- AH
- AIR T bot
- AIR T top
- AIR temp
- GPS speed kmh
- HUMIDEX SR
- IO flag
- IR back
- IR down
- IR front
- IR left
- IR right
- IR spot gnd
- IR spot left
- IR spot right
- IR spot sky
- IR up
- MRT S
- OPT
- PET SR
- SUN alt
- SUN az
- UTCI SR
- WIND AWA
- WIND TWD
- WIND TWS
- fp

Atmo raw

- left pm1
- left pm10
- left pm2.5
- right pm1
- right pm10
- right pm2.5
- left humidity
- left temperature
- right humidity
- right temperature

Atmo average

- pm1 avg
- pm10 avg
- pm2.5 avg

LYS1 raw

- LYS1 kelvin
- LYS1 lux
- LYS1 lux cat
- LYS1 b'
- LYS1 g'
- LYS1 medi
- LYS1 movement
- LYS1 r'
- LYS1 rgbb
- LYS1 rgbg
- LYS1 rgbir
- LYS1 rgbr

LYS2 raw

- LYS2 kelvin
- LYS2 lux
- LYS2 lux cat
- LYS2 b'
- LYS2 g'
- LYS2 medi
- LYS2 movement
- LYS2 r'
- LYS2 rgbb
- LYS2 rgbg
- LYS2 rgbir
- LYS2 rgbr

LYS average

- kelvin avg
- lux avg
- medi avg

GVI imagery

- GVI 50m
- GVI grass 50m
- GVI image capture date
- GVI phase mean
- GVI plant 50m
- GVI tree 50m
- Grass phase mean
- Plant phase mean
- Tree phase mean

SVF imagery

- SVF
- Building %
- Ceiling %
- Grass %
- Plant %
- Sky %
- Total Builtup %
- Total Vegetation %
- Tree %
- Wall %

Sound

- band proportion 20-200 Hz
- band proportion 200-2000 Hz
- band proportion 2000-20000 Hz
- category
- dominant frequency (Hz)
- mechanisms proportion
- mechanisms score
- octave proportion 1000 Hz
- octave proportion 125 Hz
- octave proportion 2000 Hz
- octave proportion 250 Hz
- octave proportion 31.25 Hz
- octave proportion 4000 Hz
- octave proportion 500 Hz
- octave proportion 62.5 Hz
- octave proportion 8000 Hz
- rural natural proportion
- rural natural score
- spectral centroid (Hz)
- spectral rolloff 85% (Hz)
- unidentify proportion
- urban manmade negative proportion
- urban manmade negative score
- urban manmade neutral proportion
- urban manmade neutral score
- urban manmade positive proportion
- urban manmade positive score
- vehicle proportion
- vehicle score
- wind proportion
- wind score

Solar

- BVF
- Clearness Index
- Clearness Index G0
- Clearness Index Infer
- TVF

Empatica

- acc x
- acc y
- acc z
- eda phasic
- eda scl usiemens
- eda tonic
- heart rate
- hrv fd hf
- hrv fd hfnu
- hrv fd lf
- hrv fd lf hf ratio
- hrv fd lfnu
- hrv fd total power
- hrv fd vlf
- hrv td rmssd
- hrv td sdnn
- met
- skin temperature
- vector magnitude

Eye-tracker

- blink duration s
- distance from center
- eyelid aperture avg
- fixation duration s
- fixation rate
- gaze centrality
- gaze dispersion
- gaze velocity
- pupil change rate
- pupil diameter avg
- saccade amplitude
- saccade duration s
- saccade peak velocity
- stress composite

Questionnaire

- current comfort 1 7
- environmental comfort mean 1 5
- pleasantness 1 5
- prs11 being away mean 1 5
- prs11 coherence mean 1 5
- prs11 fascination mean 1 5
- prs11 scope mean 1 5
- prs11 total mean 1 5
- space familiarity 1 5
- stai6 mean 1 5
- thermal comfort 1 5

Metadata / QC

- Date
- Date (Atmo/LYS)
- Date (UCM)
- Date (eye)
- Datetime
- GPS HDOP
- GPS alt
- GPS hdg
- GPS lat
- GPS lon
- GPS speed valid sample count
- GVI match dist m
- HRV 5min available
- HRV 5min max gap (s)
- HRV 5min removed pct
- HRV 5min segment end
- HRV 5min segment start
- HRV 5min valid rri count
- HRV 5min valid span (s)
- MAG hdg
- ParticipantID
- PhaseID
- SVF Frames In Window
- SVF Status
- age
- bike frequency
- bmi category who
- bmi kg m2
- clothing accessories
- clothing footwear
- clothing height raw
- clothing lower body
- clothing skin tone
- clothing upper body
- clothing weight raw
- commute modes
- daily walking
- end time s
- gender
- has children
- height m
- nationality
- own bike for experiment
- participant id
- pct complete
- regular physical activity
- residence type
- residential area
- response index
- segment
- site familiarity 1 10
- vision correction
- weight kg
- years in brussels

Parameters and decisions by source

1. UCM
- Retained: AIR RH, AIR T mid, HUMIDEX, MRT, PET, SND dBA, SUN Gh, UTCI, WIND AWS.
- AH: Humidity representative: ucm_AIR_RH; AH is an alternative formulation
- AIR T bot, AIR T top, AIR temp: Air-temperature representative: ucm_AIR_T_mid; alternate height/aggregate channels not co-entered
- GPS speed kmh: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- HUMIDEX SR, MRT S, PET SR, UTCI SR: Derived alternate UCM formulations (_SR/_S suffix); use the corresponding unsuffixed UCM measure
- IO flag: Discrete indoor/outdoor setting flag excluded; experimental phase protocol already stratifies outdoor travel routes.
- IR back, IR down, IR front, IR left, IR right, IR spot gnd, IR spot left, IR spot right, IR spot sky, IR up: Directional IR radiometer channels (inputs used to compute ucm_MRT [CityFeel17]); measured solar exposure representative is ucm_SUN_Gh; not co-entered as separate radiation predictors
- OPT: Derived/redundant: ucm_OPT is operative temperature, derived from air temperature and MRT [CityFeel17]
- SUN alt, SUN az: Declared Step 2 exclusion: solar altitude and solar azimuth are outside the selected exposure set
- WIND AWA, WIND TWD: Derived wind direction/angle from ucm_WIND_AWS; wind speed representative is ucm_WIND_AWS. Circular direction coding is outside the current linear wind-speed specification.
- WIND TWS: Derived true-wind speed (device correction from AWS and related fields); wind representative: ucm_WIND_AWS
- fp: The definition, units and intended interpretation of fp have not been verified against source documentation; retain in the inventory but exclude from Step 2 analytical eligibility.

2. Atmo raw
- Inputs to retained averages: left pm1, left pm10, left pm2.5, right pm1, right pm10, right pm2.5. Highlighted as source measurements; not entered alongside their averages.
- left humidity, right humidity: Raw-source inventory only: UCM ucm_AIR_RH is the declared relative-humidity measure
- left temperature, right temperature: Raw-source inventory only: UCM ucm_AIR_T_mid is the declared ambient-temperature measure

3. Atmo average
- Retained: pm1 avg, pm10 avg, pm2.5 avg.

4. LYS1 raw
- Inputs to retained averages: LYS1 kelvin, LYS1 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS1 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS1 b', LYS1 g', LYS1 r', LYS1 rgbb, LYS1 rgbg, LYS1 rgbir, LYS1 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS1 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS1 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

5. LYS2 raw
- Inputs to retained averages: LYS2 kelvin, LYS2 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS2 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS2 b', LYS2 g', LYS2 r', LYS2 rgbb, LYS2 rgbg, LYS2 rgbir, LYS2 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS2 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS2 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

6. LYS average
- Retained: kelvin avg, lux avg.
- medi avg: Derived metric estimated from LYS RGB channels, not measured spectrally; retained light measures lys_lux_avg and lys_kelvin_avg

7. GVI imagery
- Retained: GVI 50m.
- GVI grass 50m, GVI image capture date, GVI phase mean, GVI plant 50m, GVI tree 50m, Grass phase mean, Plant phase mean, Tree phase mean: Vegetation representative: gvi_50m; alternative scales/components not co-entered

8. SVF imagery
- Retained: SVF.
- Building %, Ceiling %, Grass %, Plant %, Sky %, Total Builtup %, Total Vegetation %, Tree %, Wall %: Sky openness represented by SVF; scene percentages are separate secondary constructs, not equivalent exposures

9. Sound
- band proportion 20-200 Hz, band proportion 200-2000 Hz, band proportion 2000-20000 Hz, dominant frequency (Hz), octave proportion 1000 Hz, octave proportion 125 Hz, octave proportion 2000 Hz, octave proportion 250 Hz, octave proportion 31.25 Hz, octave proportion 4000 Hz, octave proportion 500 Hz, octave proportion 62.5 Hz, octave proportion 8000 Hz, spectral centroid (Hz), spectral rolloff 85% (Hz): Not in the PCA-informed Eq2 sound set; spectral information is distinct from sound level.
- category, mechanisms score, rural natural proportion, rural natural score, unidentify proportion, urban manmade negative proportion, urban manmade negative score, urban manmade neutral proportion, urban manmade neutral score, urban manmade positive proportion, urban manmade positive score, vehicle score, wind score: Not in the PCA-informed Eq2 sound set; sound type is distinct from sound level.
- mechanisms proportion, vehicle proportion, wind proportion: Included as an additional sound main effect in Eq2 following the PCA-informed sound screen.

10. Solar
- BVF, TVF: Annular building/tree view fractions; sky openness representative is SVF; not co-entered in core Step-2 screen
- Clearness Index: Derived clearness index (kt = Gh/G0); measured solar exposure is ucm_SUN_Gh [ClearCI p.3 eq.2]
- Clearness Index G0: Calculated extraterrestrial irradiance G0 (date/time/location only), not a measured or personal exposure; denominator of Clearness_Index (kt = Gh/G0) [ClearCI p.3 eq.2]; measured solar exposure is ucm_SUN_Gh.
- Clearness Index Infer: Derived categorical clearness inference from Clearness_Index; not a continuous numeric exposure; no valid continuous-effect parameterization is declared. The original field remains in the input. Measured solar exposure is ucm_SUN_Gh

11. Empatica
- acc x, acc y, acc z, met, vector magnitude: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- eda phasic, eda tonic, heart rate, hrv fd hf, hrv fd hfnu, hrv fd lf, hrv fd lf hf ratio, hrv fd lfnu, hrv fd total power, hrv fd vlf, hrv td rmssd, hrv td sdnn, skin temperature: Used as a physiological or ocular response, not an environmental predictor.
- eda scl usiemens: Signal quality, missingness verification, or HRV window integrity metric.

12. Eye-tracker
- blink duration s, distance from center, eyelid aperture avg, fixation duration s, fixation rate, gaze centrality, gaze dispersion, gaze velocity, pupil change rate, pupil diameter avg, saccade amplitude, saccade duration s, saccade peak velocity: Used as a physiological or ocular response, not an environmental predictor.
- stress composite: Signal quality, missingness verification, or HRV window integrity metric.

13. Questionnaire
- current comfort 1 7, environmental comfort mean 1 5, pleasantness 1 5, prs11 being away mean 1 5, prs11 coherence mean 1 5, prs11 fascination mean 1 5, prs11 scope mean 1 5, prs11 total mean 1 5, space familiarity 1 5, stai6 mean 1 5, thermal comfort 1 5: Signal quality, missingness verification, or HRV window integrity metric.

14. Metadata / QC
- Date, Date (Atmo/LYS), Date (UCM), Date (eye), Datetime, GPS HDOP, GPS alt, GPS hdg, GPS lat, GPS lon, GPS speed valid sample count, GVI match dist m, MAG hdg, ParticipantID, PhaseID, pct complete: Spatiotemporal tracking, GPS positioning, or phase grouping metadata.
- HRV 5min available, HRV 5min max gap (s), HRV 5min removed pct, HRV 5min segment end, HRV 5min segment start, HRV 5min valid rri count, HRV 5min valid span (s): Signal quality, missingness verification, or HRV window integrity metric.
- SVF Frames In Window, SVF Status: SVF calibration, segmentation, run-integrity, telemetry, or manual-validation QC; not an exposure predictor.
- age, bike frequency, bmi category who, bmi kg m2, clothing accessories, clothing footwear, clothing height raw, clothing lower body, clothing skin tone, clothing upper body, clothing weight raw, commute modes, daily walking, gender, has children, height m, nationality, own bike for experiment, participant id, regular physical activity, residence type, residential area, response index, site familiarity 1 10, vision correction, weight kg, years in brussels: Baseline participant demographic characteristics or clothing insulation metadata.
- end time s, segment: YAMNet/spectral processing provenance, completeness, timing, or frequency-coverage QC; not an exposure predictor.

STEP 2
Domain Grouping (16)

Thermal	Index	Acoustic	Radiation	Pollution	Light	Greenery
AIR T mid	HUMIDEX	SND dBA	MRT	pm1 avg	lux avg	GVI 50m
AIR RH	PET		SUN Gh	pm2.5 avg	kelvin avg	
WIND AWS	UTCI		SVF	pm10 avg		

STEP 3
Equations + design VIF
(7 equations)

Eq 1: UTCI + sound + SVF + PM2.5 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 10.91
Eq 2: UTCI + sound + SVF + PM2.5 + lux + colour temperature + vehicle_proportion + wind_proportion + mechanisms_proportion + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 12.33
Eq 3: air temperature (mid) + sound + humidity + apparent wind + mean radiant temperature + SVF + PM2.5 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (air temperature (mid) x sound) + (sound x humidity) + (sound x apparent wind) + (sound x mean radiant temperature)	max VIF = 71.98
Eq 4: HUMIDEX + sound + SVF + PM2.5 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (HUMIDEX x sound)	max VIF = 7.39
Eq 5: PET + sound + SVF + PM2.5 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (PET x sound)	max VIF = 10.89
Eq 6: UTCI + sound + SVF + PM1 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 10.42
Eq 7: UTCI + sound + SVF + PM10 + lux + colour temperature + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 10.72